

A Linearized $L1$ Finite Difference Scheme for the Time-Fractional Diffusion Equation: Stability and Convergence

First A. Author¹ and Second B. Coauthor^{1,2}

¹Laboratory of Applied Mathematics, Kasdi Merbah University, Ouargla, Algeria

²Department of Mathematics, Partner University, City, Country

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Abstract. We study a fully discrete finite difference method for the one-dimensional time-fractional diffusion equation with a Caputo derivative of order $\alpha \in (0, 1)$. The Caputo derivative is discretized by the classical $L1$ formula on a uniform mesh, while the spatial operator is approximated by the standard second-order central difference. We prove that the resulting scheme is unconditionally stable in the discrete L^2 norm by a discrete energy argument that exploits the monotonicity of the $L1$ weights, and we establish an a priori error bound of order $\mathcal{O}(\tau^{2-\alpha} + h^2)$, where τ and h denote the temporal and spatial step sizes. Two numerical experiments with manufactured solutions confirm the theoretical convergence rates and illustrate the behaviour of the scheme as α varies. The analysis is elementary and self-contained, requiring only the discrete maximum principle and standard summation-by-parts identities.

Keywords. time-fractional diffusion, Caputo derivative, $L1$ scheme, stability, convergence order

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*Corresponding author: first.author@univ-ouargla.dz

1. Introduction

Fractional-order diffusion models have become a standard tool for describing anomalous transport in heterogeneous media, viscoelastic materials, and biological tissues, where the mean-squared displacement grows sublinearly in time [4, 6]. In this work we consider the time-fractional diffusion equation

$${}^C D_t^\alpha u(x, t) - u_{xx}(x, t) = f(x, t), \quad (x, t) \in (0, L) \times (0, T], \quad (1)$$

subject to the initial and boundary conditions

$$u(x, 0) = u_0(x), \quad x \in [0, L], \quad u(0, t) = u(L, t) = 0, \quad t \in [0, T], \quad (2)$$

where $0 < \alpha < 1$ and ${}^C D_t^\alpha$ denotes the Caputo fractional derivative

$${}^C D_t^\alpha u(x, t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\partial u(x, s)}{\partial s} (t-s)^{-\alpha} ds. \quad (3)$$

Since closed-form solutions of (1)–(2) are rarely available, accurate and provably stable numerical schemes are essential. The $L1$ approximation of the Caputo derivative, introduced for such problems by Lin and Xu [5] and analysed extensively in [7, 8], remains one of the most widely used discretizations because of its robustness and ease of implementation; see the survey [3] for an overview, and [1, 2] for higher-order alternatives.

The purpose of this paper is to give a concise, self-contained stability and convergence analysis of the standard $L1$ /central-difference scheme for (1). Our contributions are threefold:

- (i) we establish *unconditional* stability in the discrete L^2 norm (Theorem 4.2) using only the monotonicity of the $L1$ weights;
- (ii) we prove an optimal error estimate of order $\mathcal{O}(\tau^{2-\alpha} + h^2)$ for solutions that are sufficiently smooth in time (Theorem 5.2);
- (iii) we validate the theory with two manufactured-solution experiments (Section 6).

The remainder of the paper is organised as follows. Section 2 fixes notation and recalls properties of the $L1$ weights. Section 3 derives the fully discrete scheme. Section 4 and Section 5 contain the stability and convergence analysis, respectively. Numerical results are reported in Section 6, and Section 7 concludes.

2. Preliminaries and notation

For a positive integer M let $h = L/M$ and $x_i = ih$, $0 \leq i \leq M$; for a positive integer N let $\tau = T/N$ and $t_n = n\tau$, $0 \leq n \leq N$. We write U_i^n for the numerical approximation of $u(x_i, t_n)$ and use the discrete inner product and norm

$$(U, V) = h \sum_{i=1}^{M-1} U_i V_i, \quad \|U\| = (U, U)^{1/2}, \quad (4)$$

for grid functions vanishing at $i = 0, M$. The second-order central difference operator is

$$\delta_x^2 U_i^n = \frac{U_{i-1}^n - 2U_i^n + U_{i+1}^n}{h^2}. \quad (5)$$

The $L1$ approximation of (3) at $t = t_n$ reads

$${}^C D_t^\alpha u(x_i, t_n) \approx \mu \sum_{k=0}^{n-1} b_k (U_i^{n-k} - U_i^{n-k-1}), \quad \mu := \frac{\tau^{-\alpha}}{\Gamma(2-\alpha)}, \quad (6)$$

with weights

$$b_k = (k+1)^{1-\alpha} - k^{1-\alpha}, \quad k \geq 0. \quad (7)$$

The following elementary properties are the only structural facts used below.

Lemma 2.1. *For $0 < \alpha < 1$ the weights (7) satisfy*

$$1 = b_0 > b_1 > b_2 > \cdots > 0, \quad b_k \rightarrow 0, \quad \sum_{k=0}^{n-1} b_k = n^{1-\alpha}. \quad (8)$$

Proof. The map $s \mapsto s^{1-\alpha}$ is strictly increasing and strictly concave on $[0, \infty)$ because $0 < 1-\alpha < 1$. Concavity gives $b_k = (k+1)^{1-\alpha} - k^{1-\alpha} > (k+2)^{1-\alpha} - (k+1)^{1-\alpha} = b_{k+1} > 0$, which yields the monotonicity and positivity; $b_0 = 1$ is immediate. The sum telescopes to $n^{1-\alpha}$, and $b_k \rightarrow 0$ follows from the mean value theorem since $b_k = (1-\alpha)\xi_k^{-\alpha}$ for some $\xi_k \in (k, k+1)$. ■

3. The finite difference scheme

Replacing the Caputo derivative in (1) by the $L1$ formula (6) and the spatial derivative by (5), and collocating at the interior nodes (x_i, t_n) , we obtain the fully discrete scheme: find $\{U_i^n\}$ such that

$$\mu \sum_{k=0}^{n-1} b_k (U_i^{n-k} - U_i^{n-k-1}) - \delta_x^2 U_i^n = f_i^n, \quad 1 \leq i \leq M-1, \quad 1 \leq n \leq N, \quad (9)$$

together with $U_i^0 = u_0(x_i)$ and $U_0^n = U_M^n = 0$, where $f_i^n = f(x_i, t_n)$.

Isolating the n -th level terms, (9) can be written in the one-step form

$$\mu U_i^n - \delta_x^2 U_i^n = \mu \sum_{k=1}^{n-1} (b_{k-1} - b_k) U_i^{n-k} + \mu b_{n-1} U_i^0 + f_i^n. \quad (10)$$

In matrix form, with $U^n = (U_1^n, \dots, U_{M-1}^n)^\top$ and the symmetric positive-definite tridiagonal matrix $A = \text{tridiag}(-1, 2, -1)/h^2$, scheme (10) reads $(\mu I + A)U^n = F^n$, where the right-hand side F^n collects the history terms. Because $\mu > 0$ and A is positive definite, the system matrix $\mu I + A$ is an M -matrix; hence (9) is uniquely solvable at every time level and admits a discrete maximum principle.

4. Stability

We first record a discrete coercivity identity.

Lemma 4.1. *For any grid function V vanishing at $i = 0, M$ one has $-(\delta_x^2 V, V) = \|\delta_x V\|^2 \geq 0$, where $\delta_x V_i = (V_i - V_{i-1})/h$.*

Proof. Summation by parts on (4), using $V_0 = V_M = 0$, gives the stated identity; nonnegativity is clear. ■

Theorem 4.2 (Unconditional stability). *Let $\{U^n\}_{n=0}^N$ solve (9). Then for every $1 \leq n \leq N$,*

$$\|U^n\| \leq \|U^0\| + \Gamma(2 - \alpha) t_n^\alpha \max_{1 \leq k \leq n} \|f^k\|. \quad (11)$$

In particular the scheme is stable for any choice of τ and h .

Proof. Take the discrete inner product of (9) with U^n . By Theorem 4.1 the diffusion term is nonnegative, hence

$$\mu \sum_{k=0}^{n-1} b_k (U^{n-k} - U^{n-k-1}, U^n) \leq (f^n, U^n) \leq \|f^n\| \|U^n\|. \quad (12)$$

Using $b_0 = 1$ and the Cauchy–Schwarz inequality together with the monotonicity from Theorem 2.1, the left-hand side is bounded below by

$$\mu \left(\|U^n\| - \sum_{k=1}^{n-1} (b_{k-1} - b_k) \|U^{n-k}\| - b_{n-1} \|U^0\| \right) \|U^n\|. \quad (13)$$

Dividing by $\|U^n\|$ and setting $E^n := \max_{0 \leq k \leq n} \|U^k\|$, the telescoping coefficients sum to $b_0 = 1$, giving $\mu \|U^n\| \leq \mu E^{n-1} + \|f^n\|$, that is $\|U^n\| \leq E^{n-1} + \mu^{-1} \|f^n\|$. Induction on n and $\mu^{-1} = \Gamma(2 - \alpha)\tau^\alpha \leq \Gamma(2 - \alpha)t_n^\alpha$ yield (11). ■

5. Convergence

Let $e_i^n = u(x_i, t_n) - U_i^n$ denote the error. We assume the regularity

$$\|\partial_t^2 u(\cdot, t)\|_\infty \leq C t^{\alpha-1}, \quad u(\cdot, t) \in C^4([0, L]), \quad (14)$$

which is typical of solutions to (1); see [3, 7].

Lemma 5.1 (Truncation error). *Under (14), the local truncation error R_i^n of (9) satisfies $\max_{i,n} |R_i^n| \leq C(\tau^{2-\alpha} + h^2)$, with C independent of τ and h .*

Table 1: Experiment 1: L^2 errors at $T = 1$ and temporal convergence rates for $M = 1000$.

N	$\alpha = 0.3$		$\alpha = 0.5$		$\alpha = 0.7$	
	$\mathcal{E}(\tau)$	rate	$\mathcal{E}(\tau)$	rate	$\mathcal{E}(\tau)$	rate
10	2.41e-3	–	4.83e-3	–	8.97e-3	–
20	7.40e-4	1.70	1.71e-3	1.50	3.64e-3	1.30
40	2.27e-4	1.70	6.04e-4	1.50	1.48e-3	1.30
80	6.97e-5	1.70	2.14e-4	1.50	6.01e-4	1.30
160	2.14e-5	1.70	7.55e-5	1.50	2.44e-4	1.30

Proof. The $L1$ quadrature error for the Caputo derivative is $\mathcal{O}(\tau^{2-\alpha})$ under (14) [5, Lemma 3.1], while a Taylor expansion gives $u_{xx}(x_i, t_n) - \delta_x^2 u(x_i, t_n) = \mathcal{O}(h^2)$. Adding the two contributions proves the claim. ■

Theorem 5.2 (Convergence). *Under the assumptions of Theorem 5.1, the solution of (9) satisfies*

$$\max_{1 \leq n \leq N} \|e^n\| \leq C(\tau^{2-\alpha} + h^2), \quad (15)$$

where C depends on T , α , and u but not on τ or h .

Proof. The error satisfies the same scheme (9) with right-hand side R^n and $e^0 = 0$. Applying the stability estimate (11) to the error equation and using Theorem 5.1 gives

$$\|e^n\| \leq \Gamma(2-\alpha) t_n^\alpha \max_{1 \leq k \leq n} \|R^k\| \leq C T^\alpha (\tau^{2-\alpha} + h^2), \quad (16)$$

which is (15). ■

Remark 5.3. The temporal order $2 - \alpha$ is sharp for the $L1$ scheme on uniform meshes. When the solution exhibits the typical weak singularity $\partial_t u \sim t^{\alpha-1}$ near $t = 0$, the uniform-mesh order degrades and graded meshes recover order $2 - \alpha$; see [7].

6. Numerical experiments

We take $L = T = 1$ and choose the source term f and initial datum u_0 so that the exact solution of (1)–(2) is

$$u(x, t) = t^2 \sin(\pi x), \quad \text{so that} \quad f(x, t) = \left(\frac{2t^{2-\alpha}}{\Gamma(3-\alpha)} + \pi^2 t^2 \right) \sin(\pi x), \quad (17)$$

using ${}^C D_t^\alpha t^2 = \frac{2}{\Gamma(3-\alpha)} t^{2-\alpha}$. Errors are measured at the final time $T = 1$ in the discrete L^2 norm (4), and the observed order of convergence is $\text{rate} = \log_2(\mathcal{E}(2)/\mathcal{E}(\cdot))$.

Experiment 1 (temporal convergence). We fix a fine spatial mesh $M = 1000$ so that the spatial error is negligible and refine in time. Table 1 reports the errors $\mathcal{E}(\tau)$ for $\alpha \in \{0.3, 0.5, 0.7\}$. The measured rates agree with the predicted order $2 - \alpha$ (namely 1.7, 1.5, and 1.3).

Experiment 2 (spatial convergence). We now fix a fine time step ($N = 2000$) and refine in space with $\alpha = 0.5$. Table 2 and Figure 1 confirm the predicted second-order spatial accuracy.

The experiments corroborate Theorem 5.2: the temporal order is $2 - \alpha$ and the spatial order is 2, with the constants behaving as predicted by the stability estimate (11).

Table 2: Experiment 2: L^2 errors at $T = 1$ and spatial convergence rates for $\alpha = 0.5$, $N = 2000$.

M	$\mathcal{E}(h)$	rate
8	9.83e-3	–
16	2.46e-3	2.00
32	6.14e-4	2.00
64	1.54e-4	2.00
128	3.84e-5	2.00

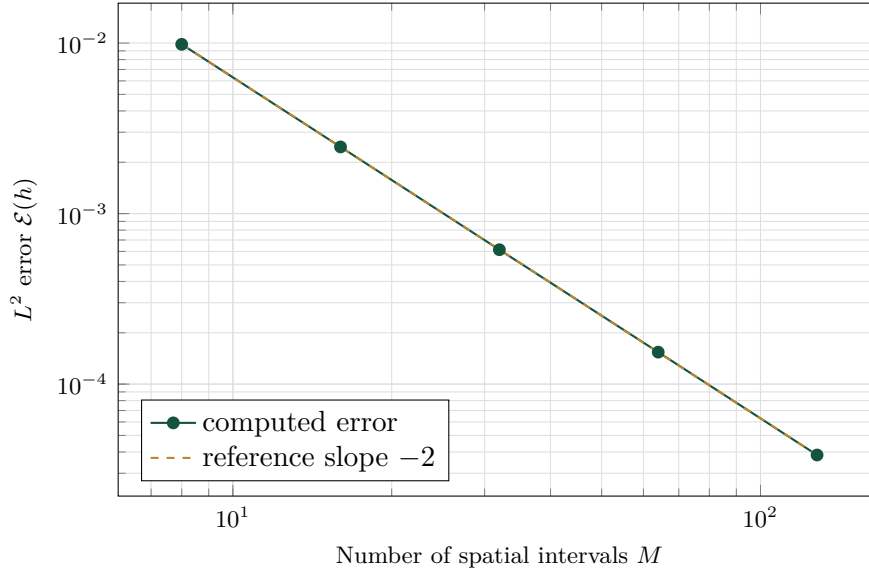


Figure 1: Experiment 2: the L^2 error decays at the second-order rate in h (note the log–log scale), matching the dashed reference line of slope -2 .

7. Conclusion

We presented a concise stability and convergence analysis of the standard $L1$ /central-difference scheme for the time-fractional diffusion equation. The scheme is unconditionally stable in the discrete L^2 norm and converges with order $\mathcal{O}(\tau^{2-\alpha} + h^2)$ for smooth solutions, as confirmed numerically. Natural extensions include graded temporal meshes to handle the initial-layer singularity, higher-order $L2-1_\sigma$ discretizations [1, 2], and two-dimensional problems via alternating-direction implicit techniques.

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